

Christian Bacalhau

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EDUCATION

Northeastern University

Boston, MA

Candidate for a Bachelor of Science degree in Computer Science | GPA: 3.6/4.0

September 2023 - May 2027

- **Concentration:** Artificial Intelligence | **Minor:** Computational Data Analytics

Relevant Coursework

Object-Oriented Design, Algorithms and Data Structures, Artificial Intelligence, Projects in Cloud Computing, Programming in C++, Discrete Structures, Mathematics of Data Models, Foundations of Cyber Security

TECHNICAL SKILLS

Languages: Java | C++ | Python | SQL | JavaScript | HTML/CSS | React | Swing | JUnit | Mockito | Pandas | C#

Developer Tools: AWS | IntelliJ | Eclipse | VS Code | Docker | Git | Datagrip | Jenkins | Azure | SQL Server

EXPERIENCE

Software Engineer

Jan. 2026 – June 2026

State Street

Burlington, MA

- Architected and developed a scalable, modular automated performance testing framework using **Jenkins**, **Python**, **Terraform**, and **Azure VMs**, enabling scheduled end-to-end performance validation for daily releases.
- Re-engineered the company's performance testing framework by migrating legacy **Groovy** tests into a new **Python** project, reducing test execution time by **32%** while simplifying the addition and maintenance of new test cases.
- Built a custom **Jenkins CI/CD** pipeline that automatically provisioned **Azure VMs**, executed distributed performance tests via the **Azure CLI**, and managed the complete testing workflow without manual intervention.
- Integrated automated reporting with **Grafana** dashboards and **SQL Server**, providing engineers with centralized visibility into performance metrics and historical test results.

Software Developer

Jan. 2025 – June 2025

Verisk

Boston, MA

- Contributed to full-stack development of a new client-facing application by designing, implementing, and testing features that enabled end-to-end user workflows.
- Built **AWS** environments for loss simulation runs, using services such as **CloudWatch**, **DynamoDB**, **Step Functions**, **Lambda**, **S3**, and **API Gateway** to support monitoring and scalability.
- Reproduced and debugged issues reported by colleagues by simulating cloud runs locally with **Docker**, streamlining the troubleshooting process and accelerating development.

PROJECTS

NU Track | VS Code, Python, SQL, Docker

January 2025

- Built a full-stack application with a team to streamline co-op and internship application tracking.
- Designed a **RESTful backend API** with modular blueprint architecture to support scalable **CRUD** operations.
- Developed a component-based front-end in **React** with reusable modules and persona-specific pages, enabling dynamic user interaction, streamlined navigation, and tailored workflows for different types of users.

Web Application | Amazon Web Services

June 2024

- Built a web application on an **EC2** instance to record student information.
- Configured an **Application Load Balancer** and **EC2** Auto Scaling for fault tolerance and high traffic handling.
- Used Amazon's **Cloud9** to simulate **600,000** requests which successfully auto-scaled and load-balanced.
- Decoupled the app with **Amazon RDS** for scalable **MySQL storage** and implemented **IAM roles** with **Secrets Manager** for secure access and communication.

Marble Solitaire | Java, Swing, JUnit, IntelliJ

June 2024

- Implemented Marble Solitaire using object-oriented programming with inheritance and encapsulation, creating a modular game structure.
- Applied **MVC pattern** by separating the model (game logic), view (user interface), and controller (user interactions) to enhance code organization and maintainability.
- Configured multiple game variants of Marble Solitaire, with a UI that lets players select their preferred version.

INTERESTS & LANGUAGES

Weightlifting | Soccer | Golf | Anime | World Travel | Proficient in Portuguese and Spanish